

HLD - Online Coding Assessment Platform

Requirements: Functional

- * Register / Login
- * View assignments
- * Start / End assessment
- * Submit code
- * View assessment status
- * View results
- * Dashboard

Non Functional requirements:

Scalable: supports thousands of students

Available: system have to work without failures.

Secure: execute code safely.

reliable: retry failed submission

Tables:

① User

- user_id
- name
- email
- password

② Assessment

- assessment_id
- title
- duration
- level

① Question

- question_id
- difficulty
- title
- explanation
- test_cases

② Submission

- submission_id
- user_id
- question_id
- code
- status

Asynchronous submission flow:

Submit code



Queue

Execute

Evaluate



Result

Status

API list:

POST /api/auth/login

GET /api/assessments

POST /api/assessments

GET /api/questions

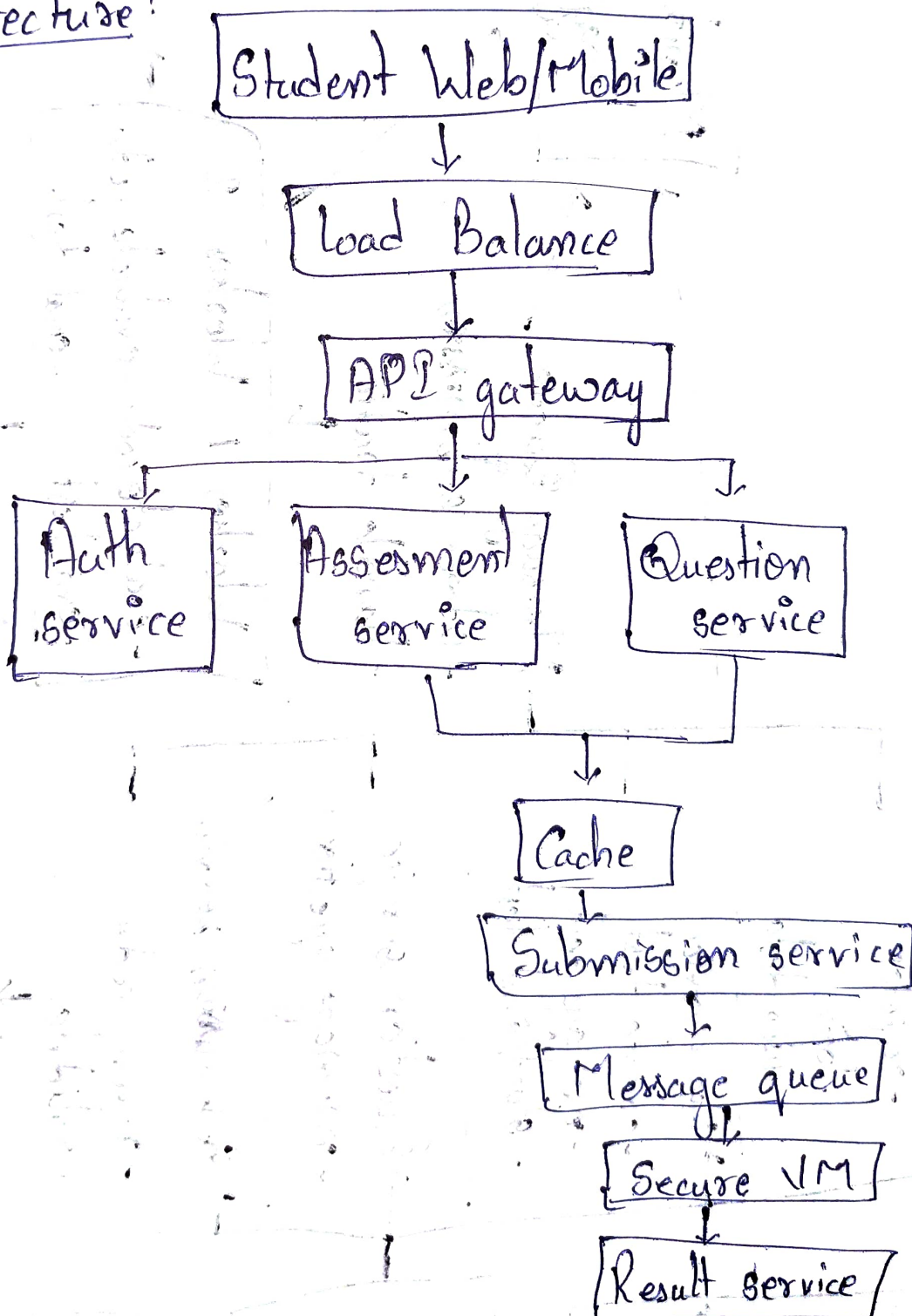
POST /api/questions

GET /api/submission/result

POST /api/submission

GET /api/submissions

Architecture:



Student
Web/Mobile

- login
- join test
- view question
- submit it
- result.

Load Balancer

- routing
- auth
- rate limit.

Admin

- create test
- add question
- view submission

Auth Service

- register
- login
- user data.

Ass. Service

- create test
- manage test
- start/end test

Question service

- manage questions

Submission service

- receive code
- validate.

Cache

- cache question
- cache user sessions.

Monitoring &
Logging

Message Queue

- stores submission
- handle traffic

Worker 1 Worker 2

- evaluate code - evaluate code
- eval. test - eval. test.

Secure VM

Database

- user
- test
- questions
- submissions
- result.

Result service

- stores result
- update status.